

# Chapter 6

## Wireless and Mobile Networks

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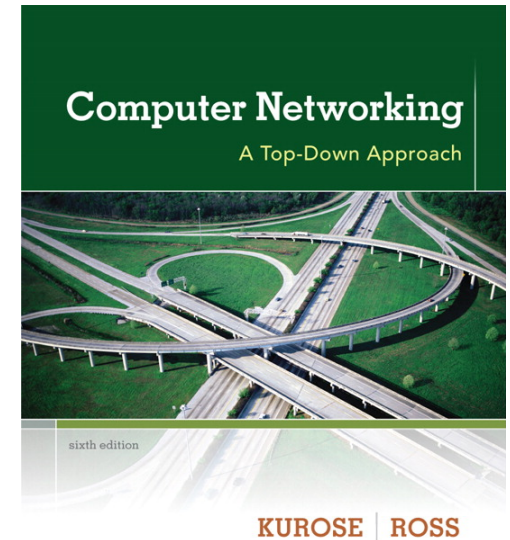
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## Computer Networking: A Top Down Approach

6<sup>th</sup> edition

Jim Kurose, Keith Ross  
Addison-Wesley

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# Ch. 6: Wireless and Mobile Networks

## Background:

- ❖ # wireless (mobile) phone subscribers now exceeds # wired phone subscribers (5-to-1)!
- ❖ # wireless Internet-connected devices equals # wireline Internet-connected devices
  - laptops, Internet-enabled phones promise anytime untethered Internet access
- ❖ two important (but different) challenges
  - *wireless*: communication over wireless link
  - *mobility*: handling the mobile user who changes point of attachment to network

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## 6.1 Introduction

### Wireless

#### 6.2 Wireless links, characteristics

- CDMA

#### 6.3 IEEE 802.11 wireless LANs (“Wi-Fi”)

#### 6.4 Cellular Internet Access

- architecture
- standards (e.g., GSM)

### Mobility

#### 6.5 Principles: addressing and routing to mobile users

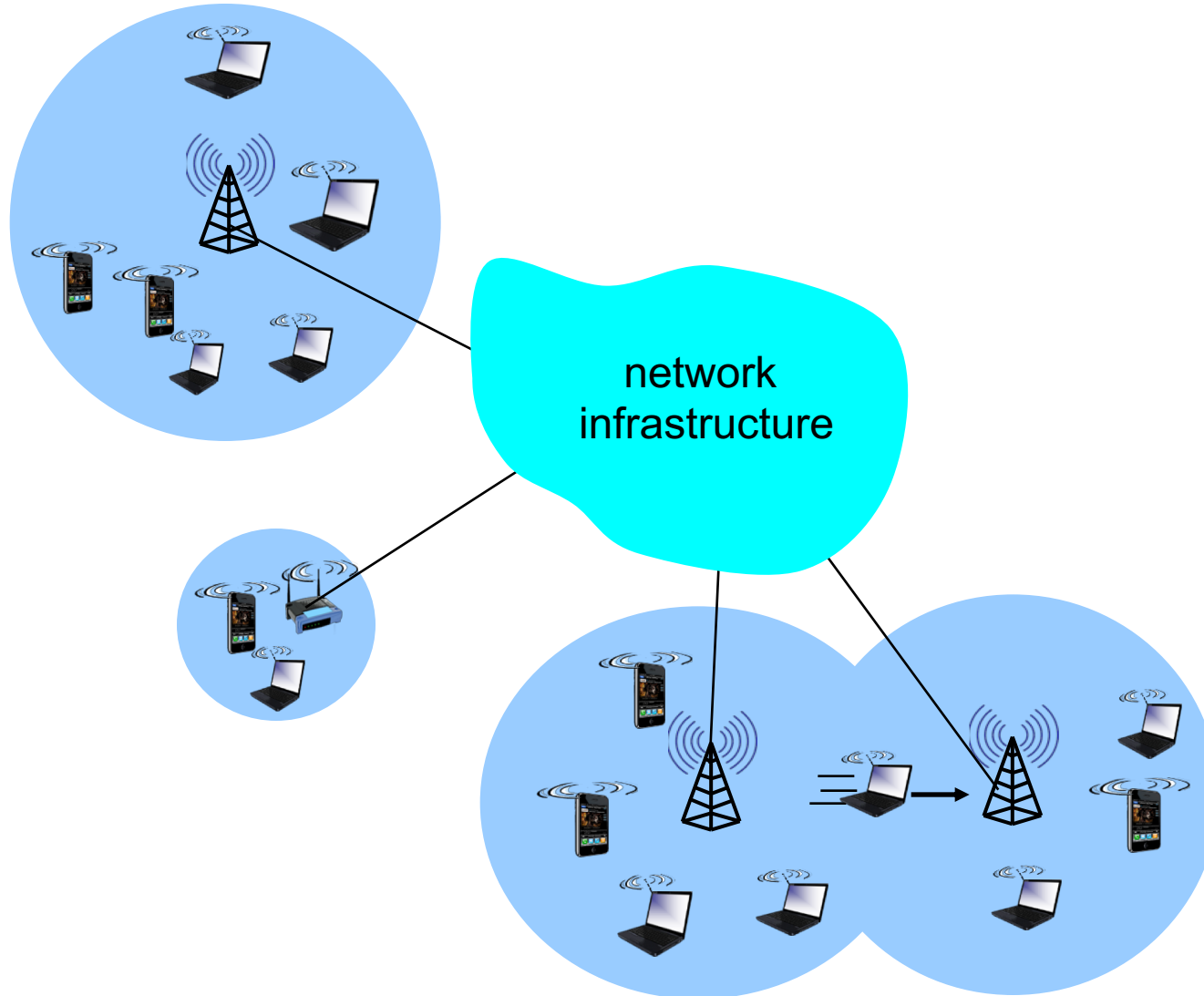
#### 6.6 Mobile IP

#### 6.7 Handling mobility in cellular networks

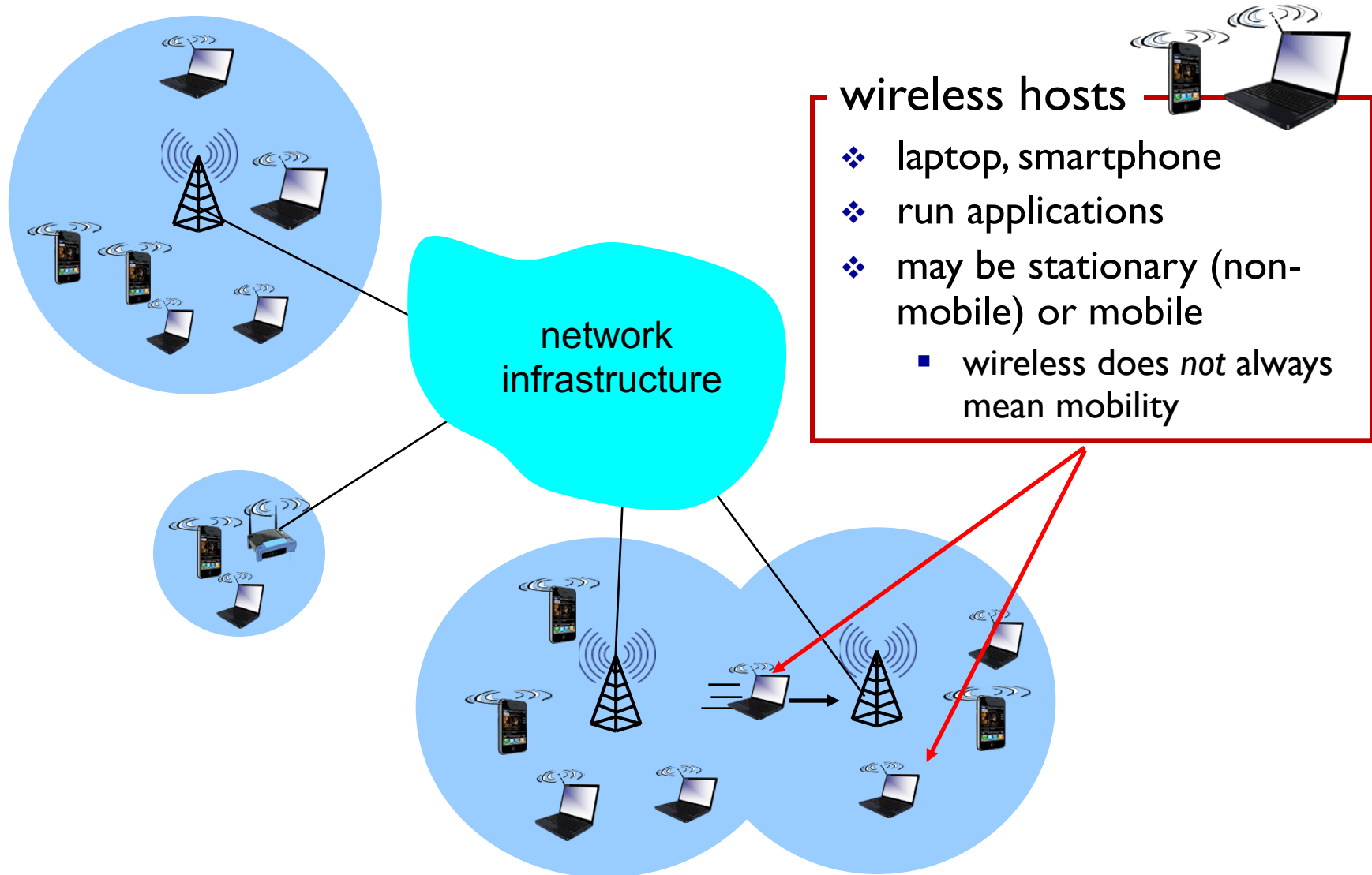
#### 6.8 Mobility and higher-layer protocols

#### 6.9 Summary

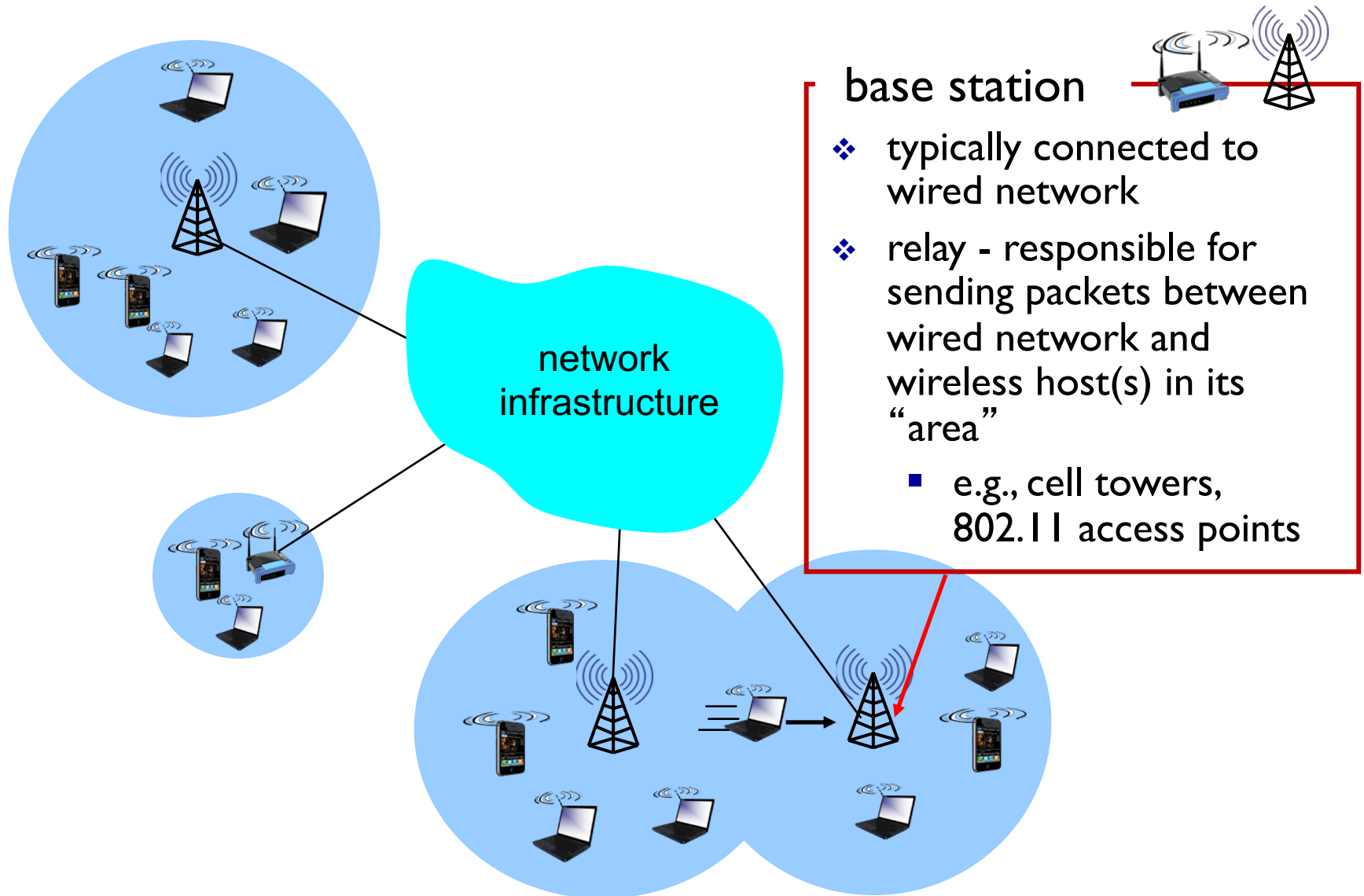
# Elements of a wireless network



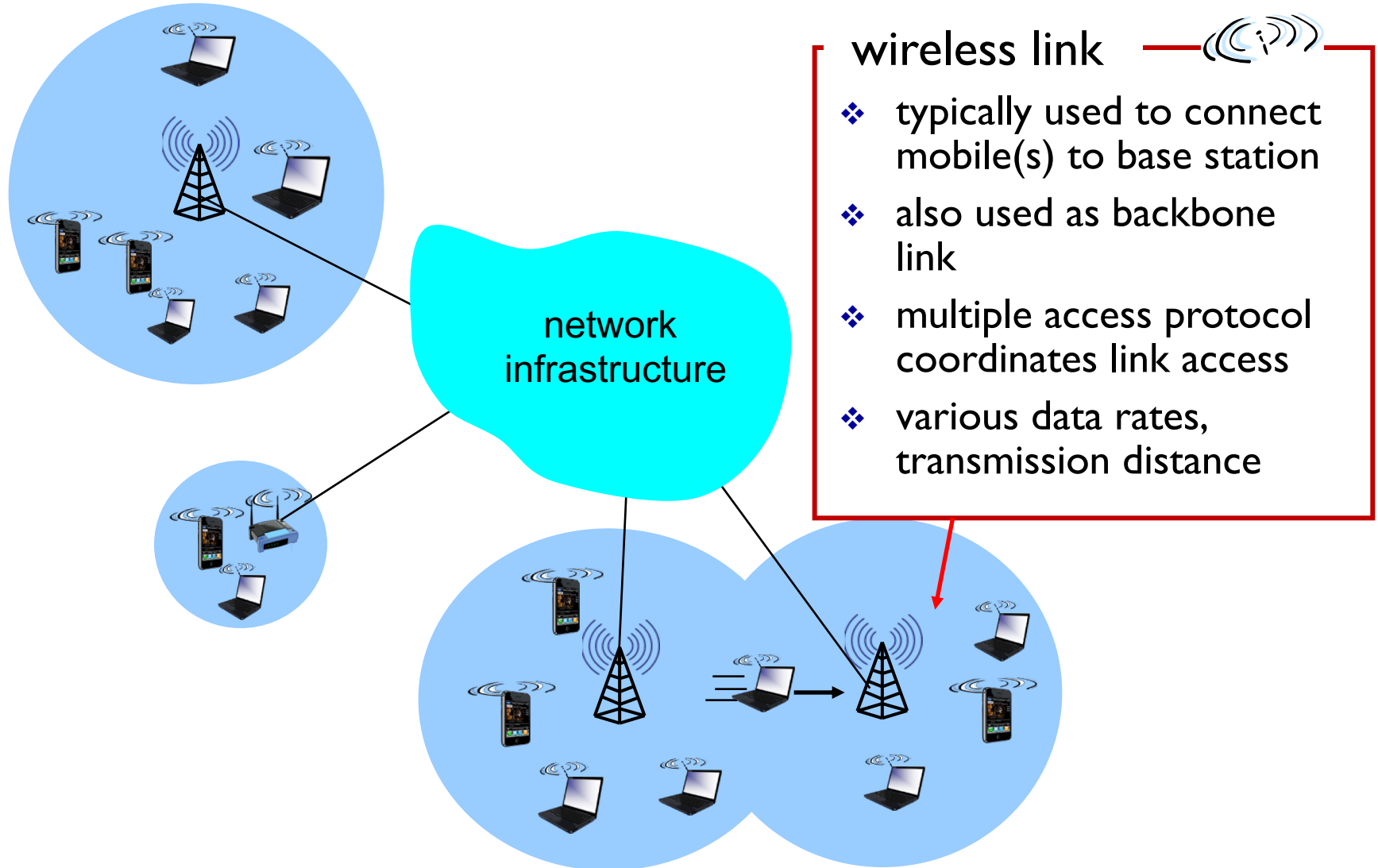
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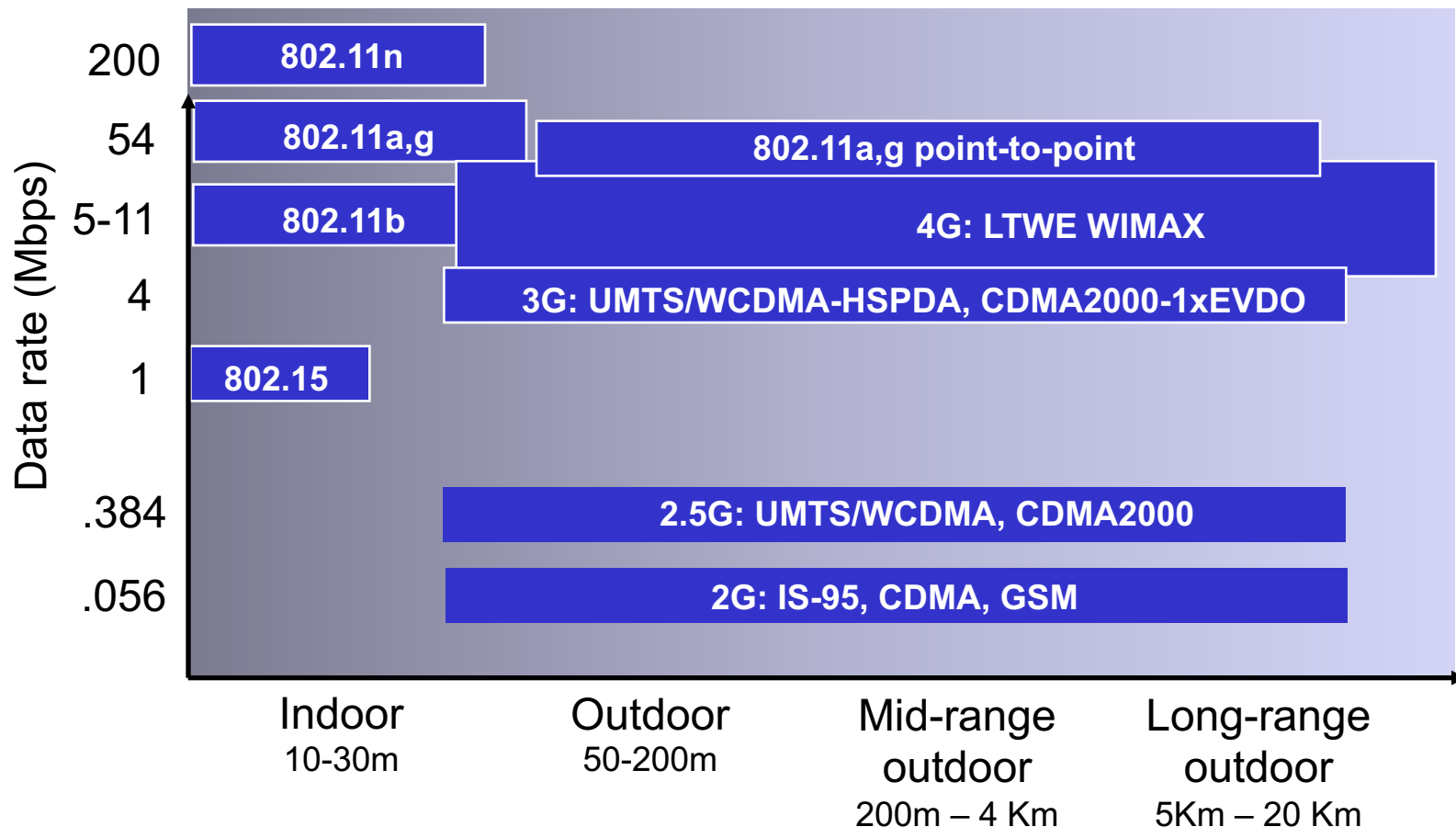
# Elements of a wireless network



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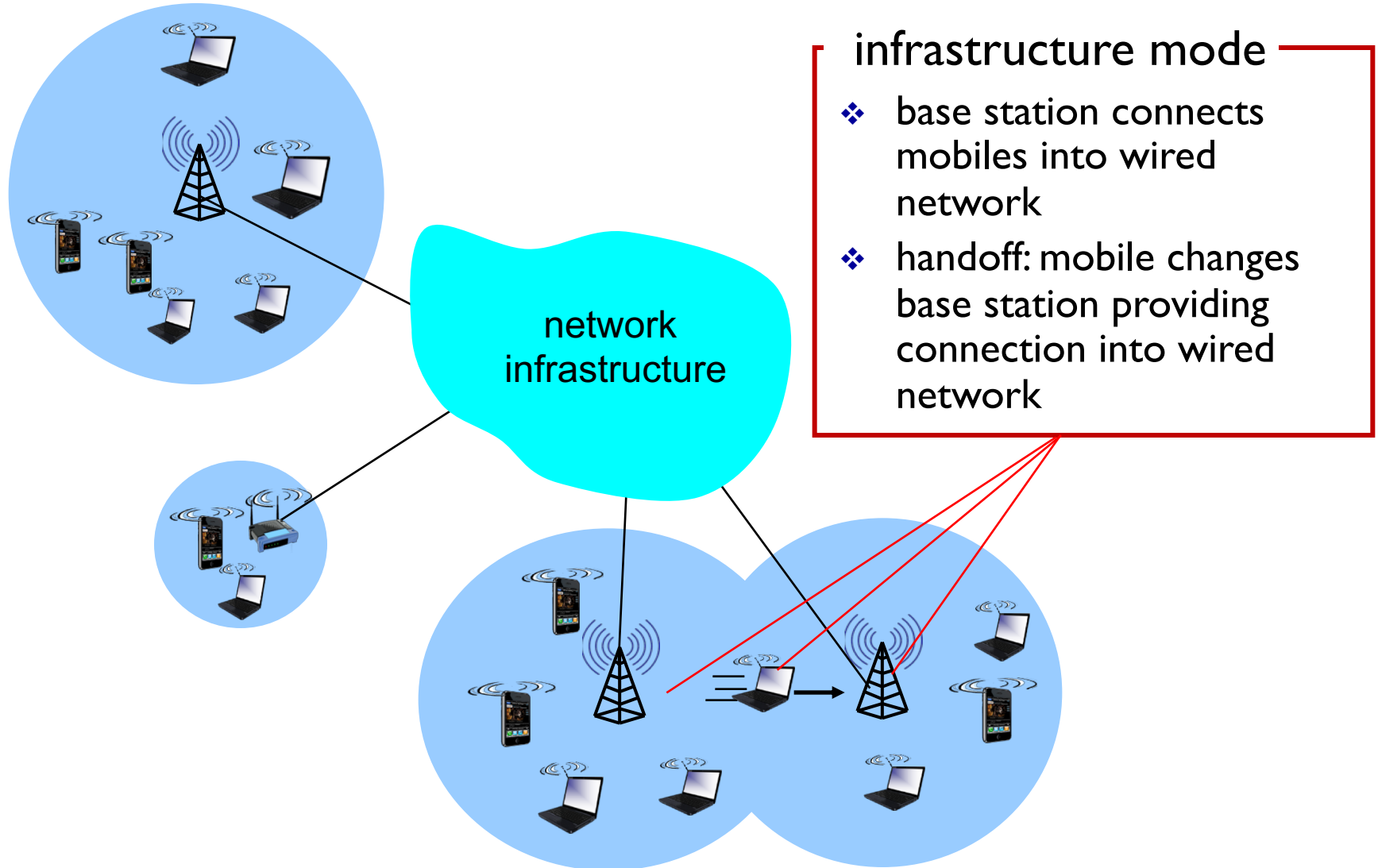


# Characteristics of selected wireless links

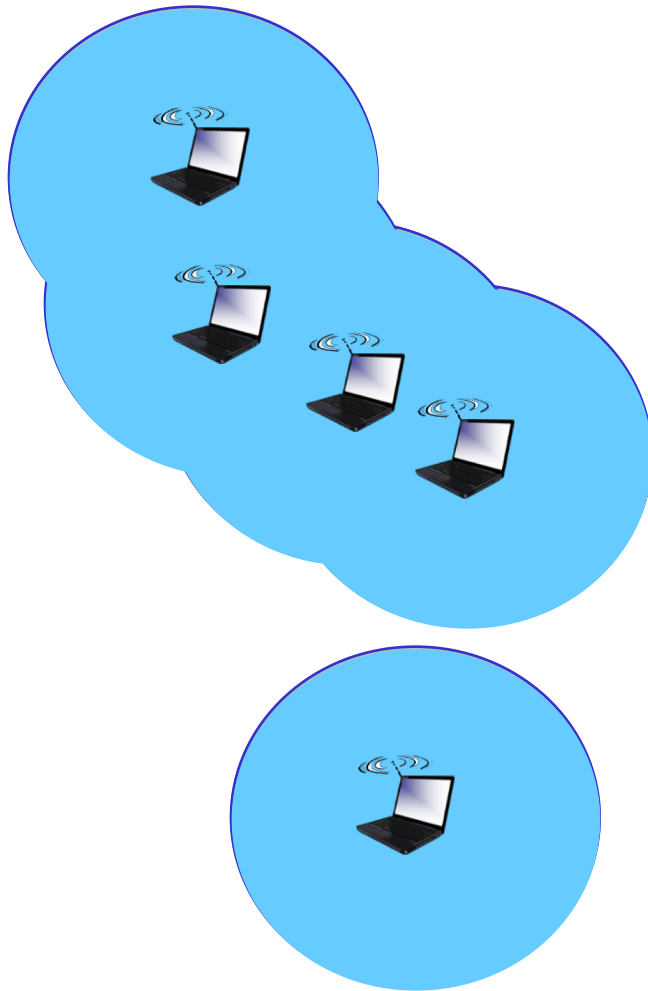




# Elements of a wireless network



# Elements of a wireless network



## ad hoc mode

- ❖ no base stations
- ❖ nodes can only transmit to other nodes within link coverage
- ❖ nodes organize themselves into a network: route among themselves

# Wireless network taxonomy

|                               | single hop                                                                                          | multiple hops                                                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| infrastructure<br>(e.g., APs) | host connects to<br>base station (WiFi,<br>WiMAX, cellular)<br>which connects to<br>larger Internet | host may have to<br>relay through several<br>wireless nodes to<br>connect to larger<br>Internet: <i>mesh net</i>                      |
| no<br>infrastructure          | no base station, no<br>connection to larger<br>Internet (Bluetooth,<br>ad hoc nets)                 | no base station, no<br>connection to larger<br>Internet. May have to<br>relay to reach other<br>a given wireless node<br>MANET, VANET |

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# Wireless Link Characteristics (I)

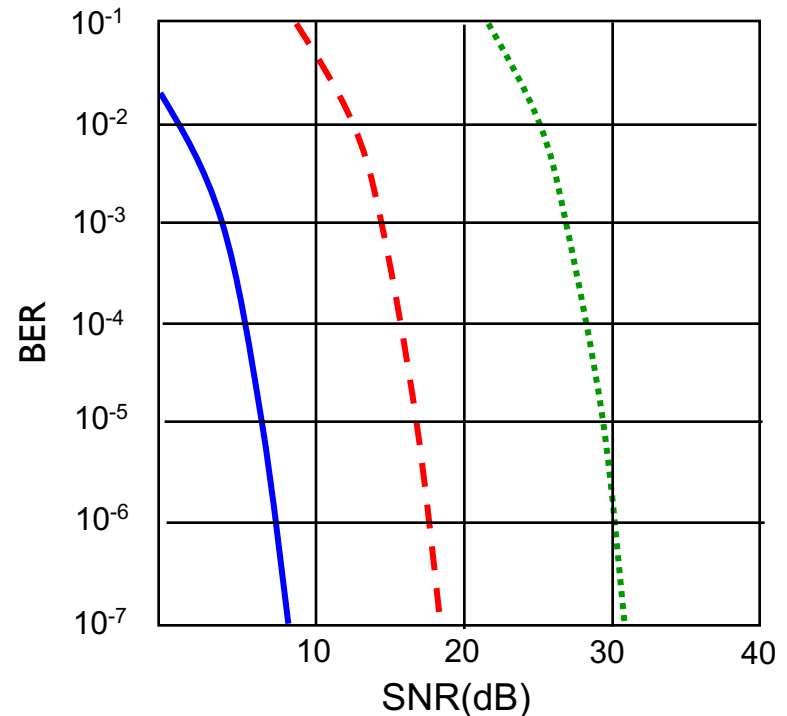
*important* differences from wired link ....

- *decreased signal strength*: radio signal attenuates as it propagates through matter (path loss)
- *interference from other sources*: standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors) interfere as well
- *multipath propagation*: radio signal reflects off objects ground, arriving at destination at slightly different times

.... make communication across (even a point to point) wireless link much more “difficult”

# Wireless Link Characteristics (2)

- ❖ SNR: signal-to-noise ratio
  - larger SNR – easier to extract signal from noise (a “good thing”)
- ❖ *SNR versus BER tradeoffs*
  - *given physical layer*: increase power  $\rightarrow$  increase SNR  $\rightarrow$  decrease BER
  - *given SNR*: choose physical layer that meets BER requirement, giving highest throughput
    - SNR may change with mobility: dynamically adapt physical layer (modulation technique, rate)



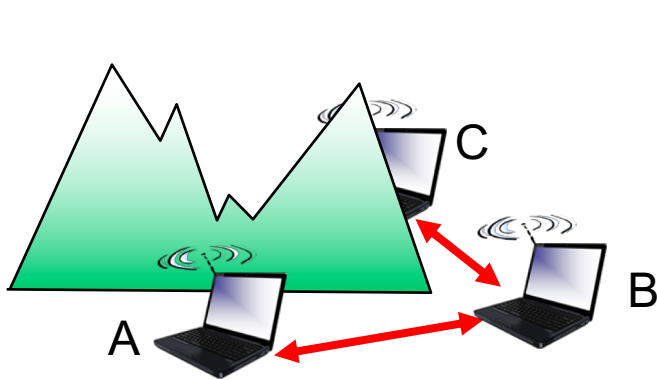
..... QAM256 (8 Mbps)

- - - QAM16 (4 Mbps)

— BPSK (1 Mbps)

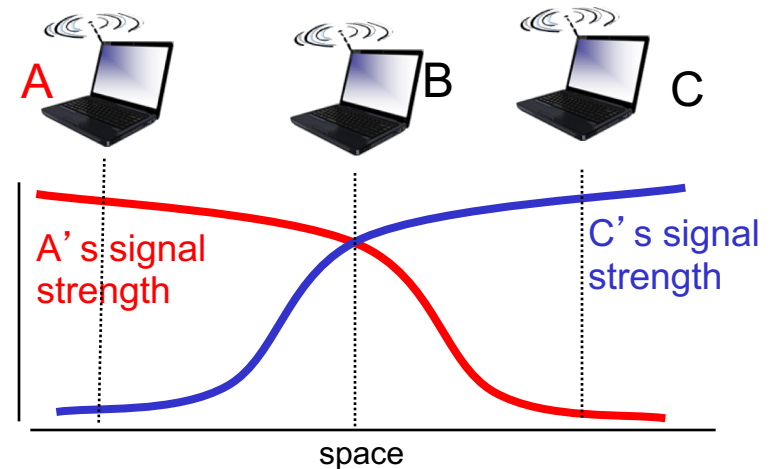
# Wireless network characteristics

Multiple wireless senders and receivers create additional problems (beyond multiple access):



## *Hidden terminal problem*

- ❖ B, A hear each other
- ❖ B, C hear each other
- ❖ A, C can not hear each other means A, C unaware of their interference at B



## *Signal attenuation:*

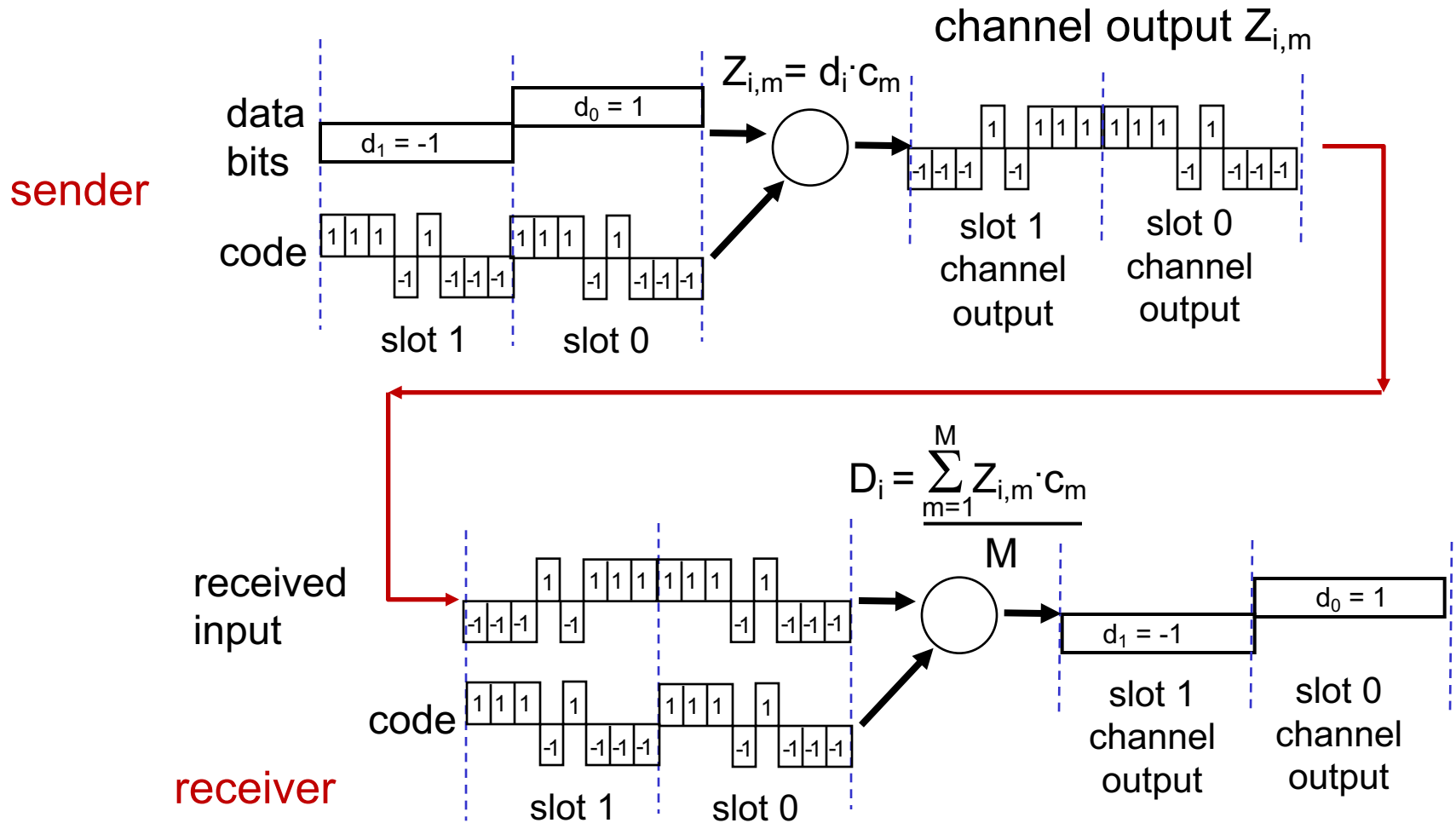
- ❖ B, A hear each other
- ❖ B, C hear each other
- ❖ A, C can not hear each other interfering at B

# Code Division Multiple Access (CDMA)

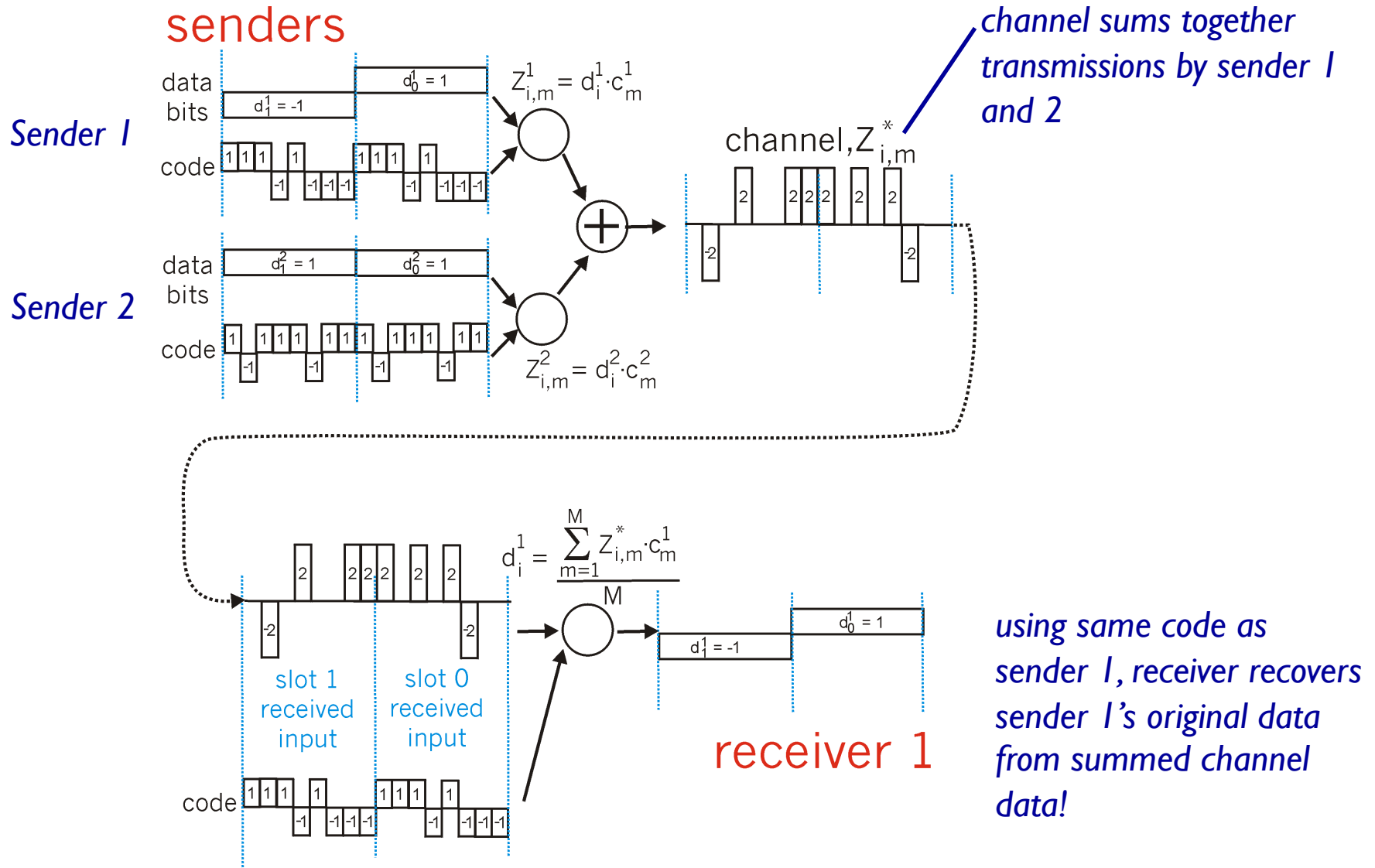
- ❖ unique “code” assigned to each user; i.e., code set partitioning
  - all users share same frequency, but each user has own “chipping” sequence (i.e., code) to encode data
  - allows multiple users to “coexist” and transmit simultaneously with minimal interference (if codes are “orthogonal”)
- ❖ *encoded signal* = (original data)  $\times$  (chipping sequence)
- ❖ *decoding*: inner-product of encoded signal and chipping sequence



# CDMA encode/decode



# CDMA: two-sender interference



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# IEEE 802.11 Wireless LAN

## 802.11b

- ❖ 2.4-5 GHz unlicensed spectrum
- ❖ up to 11 Mbps
- ❖ direct sequence spread spectrum (DSSS) in physical layer
  - all hosts use same chipping code

## 802.11a

- 5-6 GHz range
- up to 54 Mbps

## 802.11g

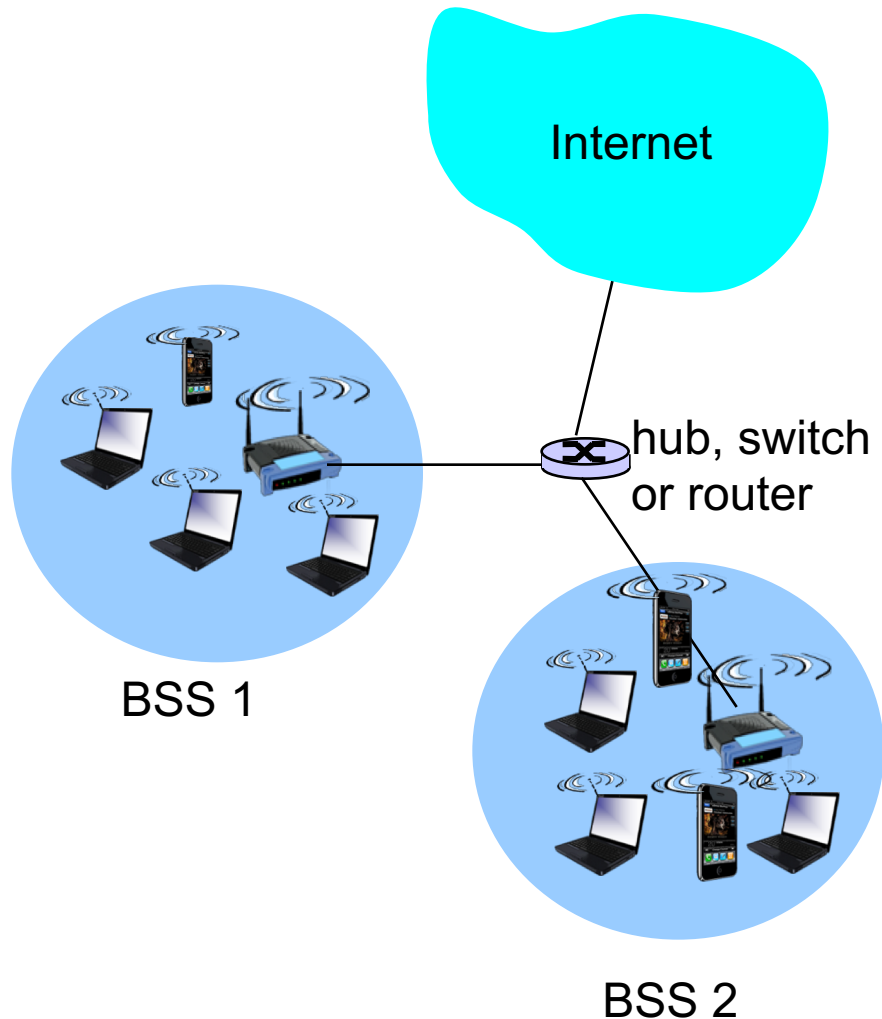
- 2.4-5 GHz range
- up to 54 Mbps

## 802.11n: multiple antennae

- 2.4-5 GHz range
- up to 200 Mbps

- 
- ❖ all use CSMA/CA for multiple access
  - ❖ all have base-station and ad-hoc network versions

# 802.11 LAN architecture

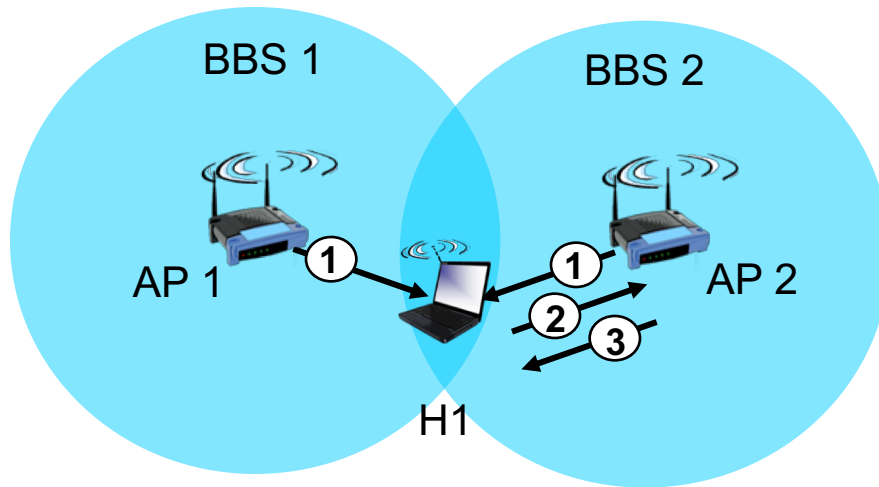


- ❖ wireless host communicates with base station
  - base station = access point (AP)
- ❖ **Basic Service Set (BSS)** (aka “cell”) in infrastructure mode contains:
  - wireless hosts
  - access point (AP): base station
  - ad hoc mode: hosts only

# 802.11: Channels, association

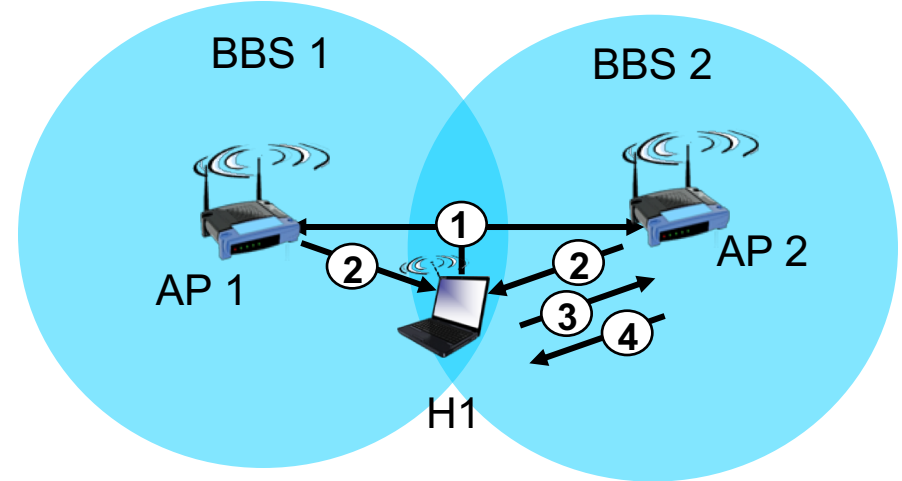
- ❖ 802.11b: 2.4GHz-2.485GHz spectrum divided into 11 channels at different frequencies
  - AP admin chooses frequency for AP
  - interference possible: channel can be same as that chosen by neighboring AP!
- ❖ host: must *associate* with an AP
  - scans channels, listening for *beacon frames* containing AP's name (SSID) and MAC address
  - selects AP to associate with
  - may perform authentication [Chapter 8]
  - will typically run DHCP to get IP address in AP's subnet

# 802.11: passive/active scanning



## passive scanning:

- (1) beacon frames sent from APs
- (2) association Request frame sent: H1 to selected AP
- (3) association Response frame sent from selected AP to H1

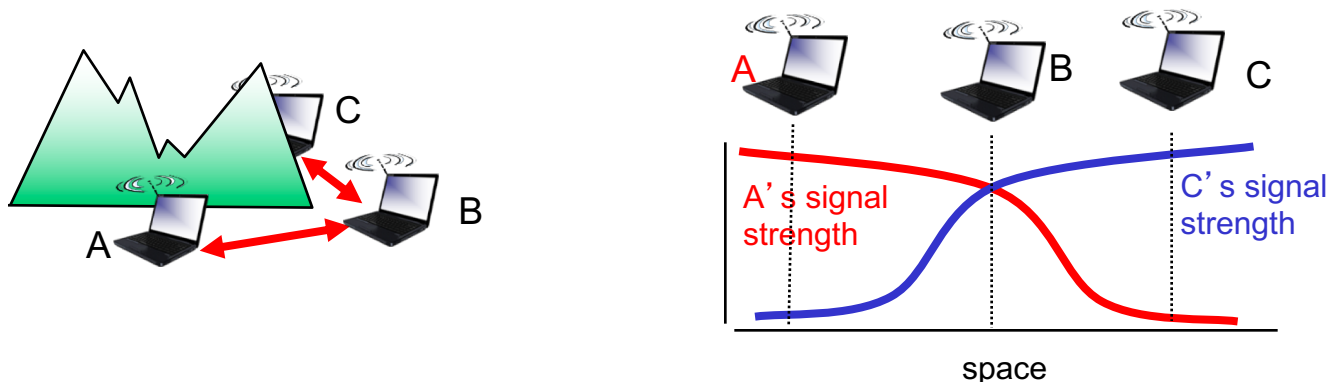


## active scanning:

- (1) Probe Request frame broadcast from H1
- (2) Probe Response frames sent from APs
- (3) Association Request frame sent: H1 to selected AP
- (4) Association Response frame sent from selected AP to H1

# IEEE 802.11: multiple access

- ❖ avoid collisions: 2<sup>+</sup> nodes transmitting at same time
- ❖ 802.11: CSMA - sense before transmitting
  - don't collide with ongoing transmission by other node
- ❖ 802.11: *no* collision detection!
  - difficult to receive (sense collisions) when transmitting due to weak received signals (fading)
  - can't sense all collisions in any case: hidden terminal, fading
  - goal: *avoid collisions*: CSMA/C(ollision)A(voidance)





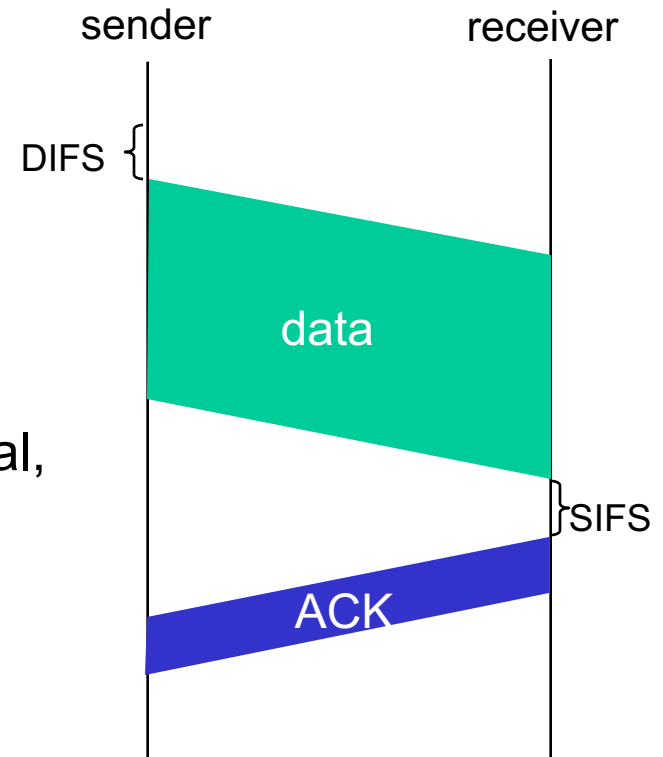
# IEEE 802.11 MAC Protocol: CSMA/CA

## 802.11 sender

- 1 if sense channel idle for **DIFS** then  
transmit entire frame (no CD)
- 2 if sense channel busy then  
start random backoff time  
timer counts down while channel idle  
transmit when timer expires  
if no ACK, increase random backoff interval,  
repeat 2

## 802.11 receiver

- if frame received OK  
return ACK after **SIFS** (ACK needed due to  
hidden terminal problem)



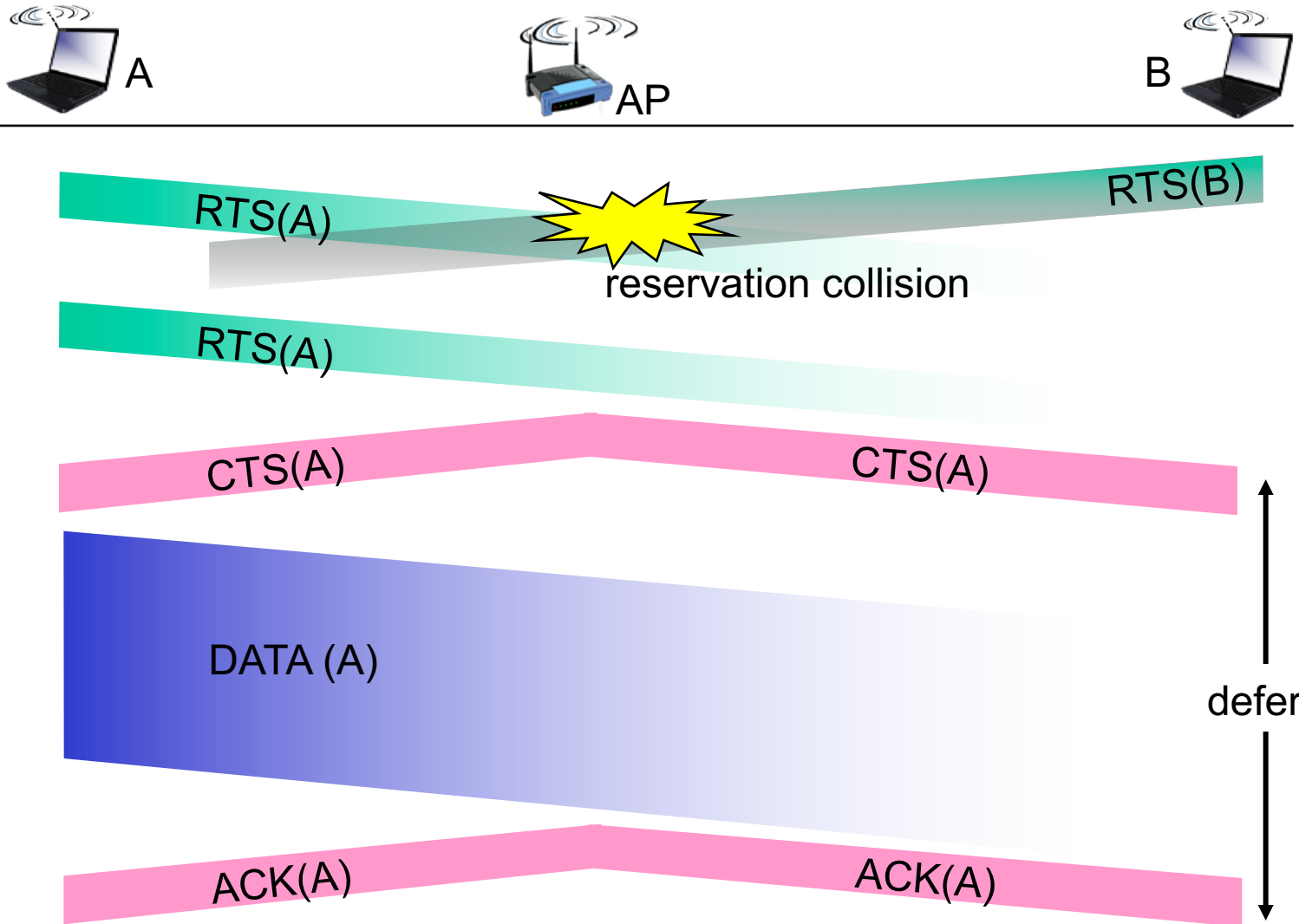
# Avoiding collisions (more)

*idea:* allow sender to “reserve” channel rather than random access of data frames: avoid collisions of long data frames

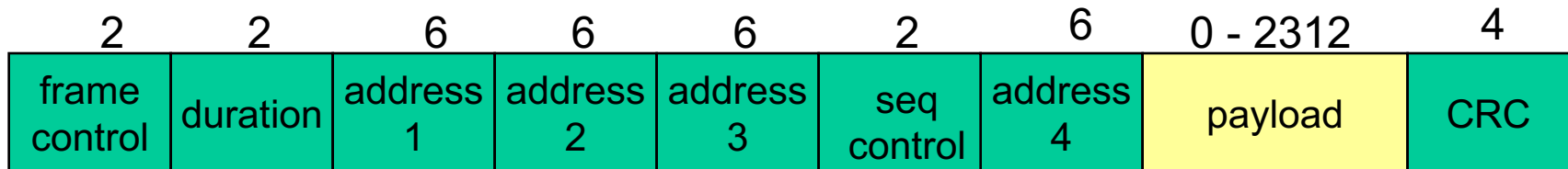
- ❖ sender first transmits *small* request-to-send (RTS) packets to BS using CSMA
  - RTSs may still collide with each other (but they’re short)
- ❖ BS broadcasts clear-to-send CTS in response to RTS
- ❖ CTS heard by all nodes
  - sender transmits data frame
  - other stations defer transmissions

*avoid data frame collisions completely  
using small reservation packets!*

# Collision Avoidance: RTS-CTS exchange



# 802.11 frame: addressing



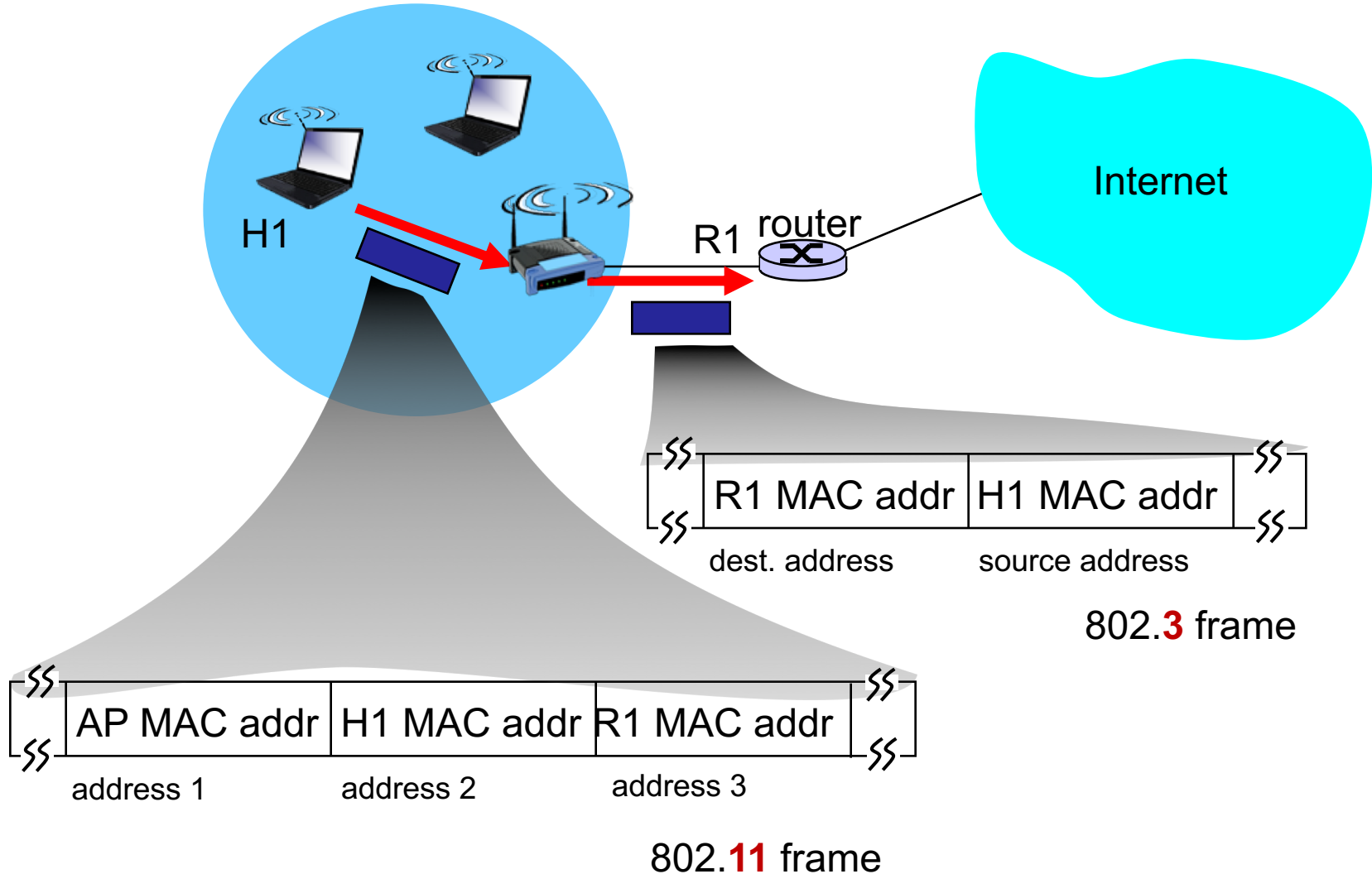
**Address 1:** MAC address of wireless host or AP to receive this frame

**Address 2:** MAC address of wireless host or AP transmitting this frame

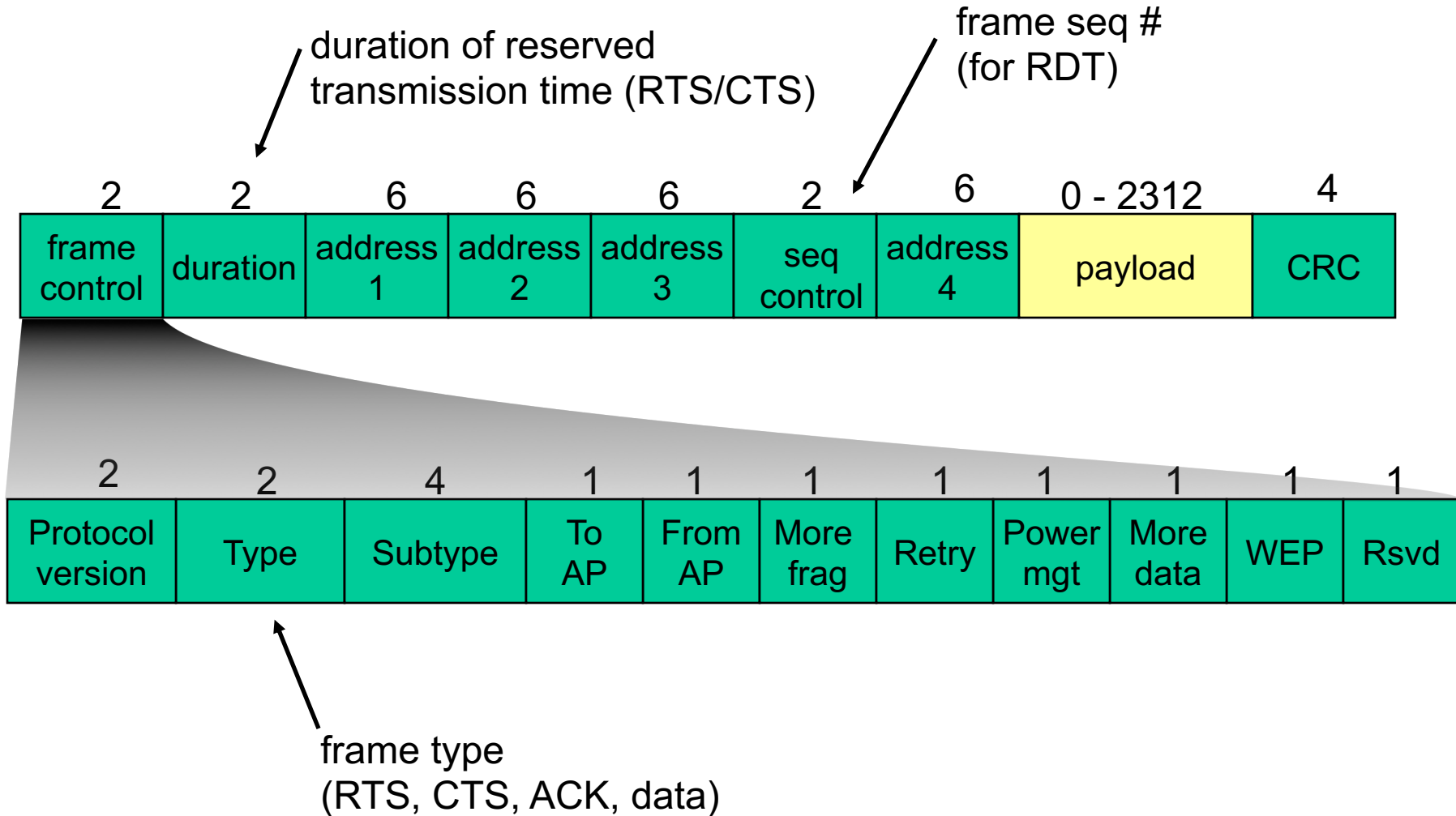
**Address 3:** MAC address of router interface to which AP is attached

**Address 4:** used only in ad hoc mode

# 802.11 frame: addressing

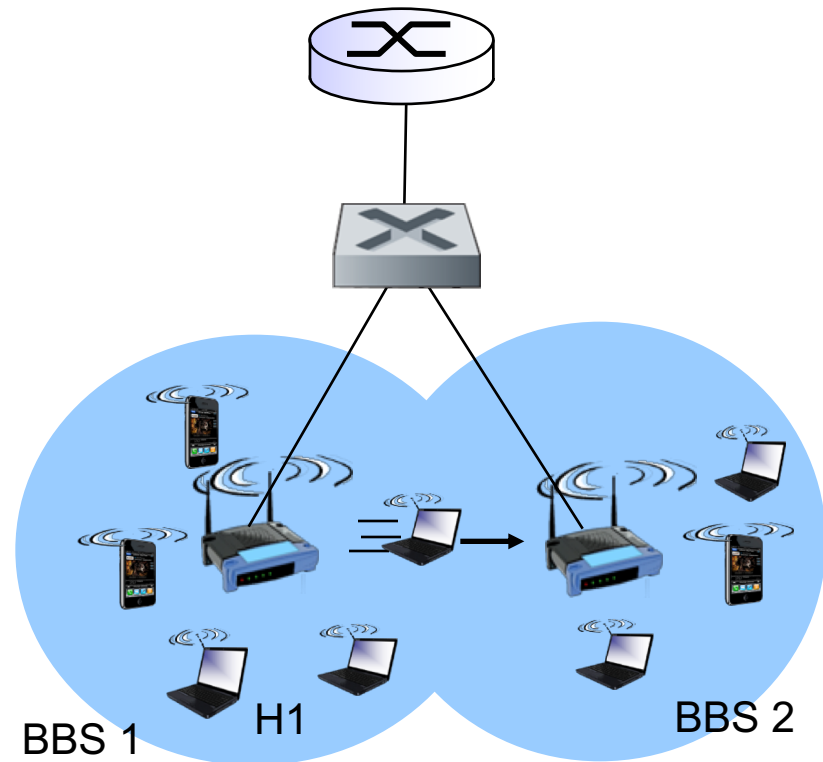


# 802.11 frame: more



# 802.11: mobility within same subnet

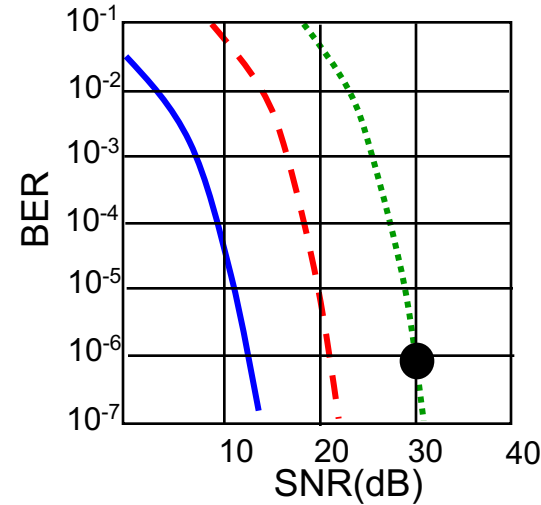
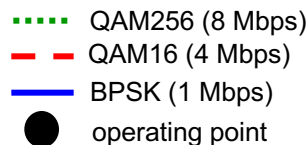
- ❖ HI remains in same IP subnet: IP address can remain same
- ❖ switch: which AP is associated with HI?
  - self-learning (Ch. 5): switch will see frame from HI and “remember” which switch port can be used to reach HI



# 802.11: advanced capabilities

## *Rate adaptation*

- ❖ base station, mobile dynamically change transmission rate (physical layer modulation technique) as mobile moves, SNR varies



1. SNR decreases, BER increase as node moves away from base station
2. When BER becomes too high, switch to lower transmission rate but with lower BER



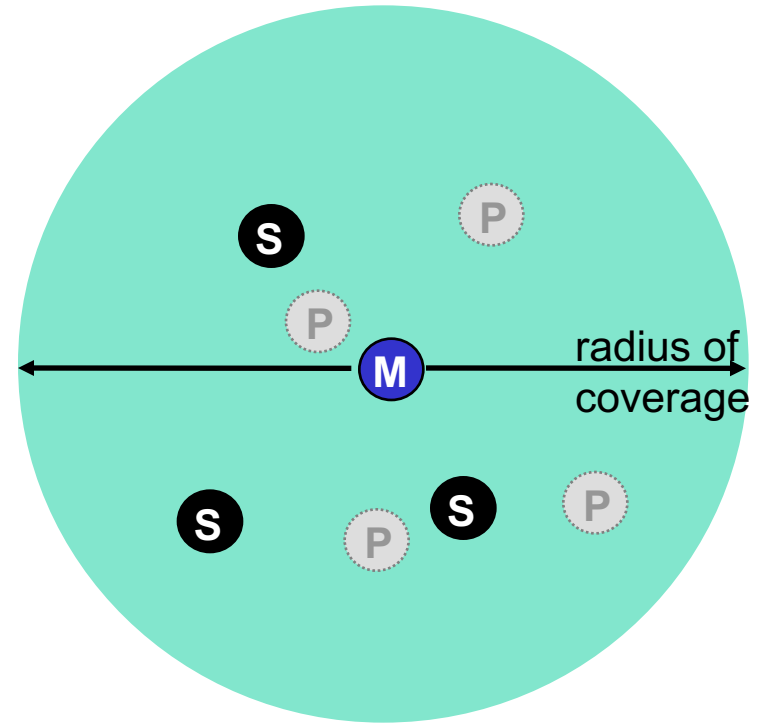
# 802.11: advanced capabilities

## *power management*

- ❖ node-to-AP: “I am going to sleep until next beacon frame”
  - AP knows not to transmit frames to this node
  - node wakes up before next beacon frame
- ❖ beacon frame: contains list of mobiles with AP-to-mobile frames waiting to be sent
  - node will stay awake if AP-to-mobile frames to be sent; otherwise sleep again until next beacon frame

# 802.15: personal area network

- ❖ less than 10 m diameter
- ❖ replacement for cables (mouse, keyboard, headphones)
- ❖ ad hoc: no infrastructure
- ❖ master/slaves:
  - slaves request permission to send (to master)
  - master grants requests
- ❖ 802.15: evolved from Bluetooth specification
  - 2.4-2.5 GHz radio band
  - up to 721 kbps



- M** Master device
- S** Slave device
- P** Parked device (inactive)

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